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INTERIM REVISION OF
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INTERIM FEDERAL SPECIFICATION

CHALK, CARPENTERS' AND RAILROAD

This Interim Federal Specification was developed by the Standardization Division, Federal Supply Service, General Services Administration, Washington, D.C. 20406, based upon currently available technical information. It is recommended that Federal agencies use it in procurement and forward recommendations for changes to the preparing activity at the address shown above.

The General Services Administration has authorized the use of this Interim Federal Specification as a valid exception to Federal Specification SS-C-255, dated August 30, 1949.

1. SCOPE AND CLASSIFICATION.

1.1 Scope. This specification covers two types of chalk, one type suitable for carpenters' use and the other type suitable for railroad use (see 6.1).

1.2 Classification.

1.2.1 Types. The chalk shall be of the following types, as specified (see 6.2):

Type I - Carpenters' chalk.
Type II- Railroad chalk.

1.2.2 Color. The chalk shall be white, blue, red, or yellow, as specified (see 6.2).

2. APPLICABLE DOCUMENTS

2.1 Specifications and standards. The following specifications and standards, of the issues in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

Federal Specifications:

PPP-B-585 - Boxes, Wood, Wirebound.
PPP-B-591 - Boxes, Fiberboard, Wood-cleated.
PPP-B-601 - Boxes, Wood, Cleated-Plywood.
PPP-B-621 - Boxes, Wood, Nailed and Lock-Corner.
PPP-B-636 - Box, Fiberboard.
PPP-B-640 - Boxes, Fiberboard, Corrugated, Triple-Wall.
PPP-B-665 - Boxes, Paperboard, Metal Stayed (Including Stay Material).
PPP-B-676 - Boxes, Setup.

Federal Standard :

Fed. Std. No. 123 - Marking for Domestic Shipment (Civilian Agencies).

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(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, D.C. 20402.

(Single copies of this specification and other Federal Specifications required by activities outside the Federal Government for bidding purposes are available without charge from Business Service Centers at the General Services Administration Regional Offices in Boston, New York, Washington, D.C., Atlanta, Chicago, Kansas City, Mo., Fort Worth, Denver, San Francisco, Los Angeles, and Seattle, Washington.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

Military Standards:

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.

(Copies of Military Specifications and Standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following document forms a part of the specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

American Society for Testing and Materials Publication:

ASTM Designation: C 471-61 Chemical Analysis of Gypsum and Gypsum Products.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, Pennsylvania 10103.)

3. REQUIREMENTS

3.1 Materials. All materials used in the composition of the chalk shall be free from grit, hard spots, and sandy abrasives.

3.2 Composition.

3.2.1 White chalk. White chalk shall contain not less than 90 percent, by weight, calcium, sulfate ($CaSO_4 2H_2O$) when tested as specified in 4.2.3.1. The remainder shall consist of suitable binder necessary to control the degree of hardness.

3.2.2 Colored chalk. Colored chalk shall contain not less than 85 percent, by weight, calcium sulfate ($CaSO_4 2H_2O$) when tested as specified in 4.2.3.1. The remainder shall consist of colored pigment and suitable binder necessary to control the degree of hardness.

3.3 Size and shape.

3.3.1 Type I. The carpenters' chalk shall be hemispherical in shape, shall have a base diameter of not less than 2 inches, and shall be not less than 3/4 inch in height.

3.3.2 Type II. Railroad chalk shall be cylindrical in shape, shall be not less than 3-15/16 inches long, and shall have a diameter of 1 inch $\pm$ I/16 inch at the large end, tapering to 7/8 inch $\pm$ 1/16 inch at the opposite end. The smaller end shall be rounded.

3.4 Hardness.

3.4.1 Type I. The hardness shall be of such degree as to permit the chalk to transfer freely to a string (chalk line) drawn across its surface, and to leave the string thus making a smooth, uniform, legible mark against a smooth surface such as steel, wood, concrete, or other similar surfaces in construction work.

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3.4.2 Type II. The hardness shall be of such degree as to permit the chalk to make a smooth, uniform, legible mark on either wood, steel, rough metal, concrete, or other construction materials not having a glossy surface.

3.5 Weight.

3.5.1 Type I. The carpenters' chalk shall weigh not less than 8 pounds a gross.

3.5.2 Type II. The railroad chalk shall weigh not less than 10 pounds a gross.

3.6 Breaking strength. The chalk shall not break or crack apart when tested as required in 4.2.3.2 or 4.2.3.3, as applicable.

3.7 Coating. When specified (see 6.2), type II chalk shall be uniformly coated with a suitable enamel for cleanliness in handling.

3.8 Workmanship. The chalk shall have a smooth, uniform surface and shall be free from holes, cracks, ripples, lumps, excessive dust, discoloration, foreign matter, or any defect which may cause undesirable effects during use.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein.. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.2 Sampling and inspection.

4.2.1 Inspection of end item. Sample chalk shall be examined for defects listed in table I.. Sampling shall be in accordance with MIL-STD-105, inspection level S-2, with an acceptable quality level (AQL) of 4.0 percent defective. The lot size shall consist of all packages containing one type and color of chalk offered for acceptance at one time. The sample unit (unit of product) shall be one package. One or more defects to a package shall constitute a defective package.

TABLE I. Inspection for end item defects.

Examine	Defects
Type and color	Not as specified (see 6.2)
Material	Gritty.
Size and shape	Not as specified in 3.3.1 or 3.3.2, as applicable.
Hardness	Not as specified in 3.4
Weight	Type I carpenters' chalk weighs less than 8 pounds a gross, Type II railroad chalk weighs less than 10 pounds a gross.
Coating	Not furnished when required (see 6.2).
Workmanship	Surface not smooth and uniform; not free from holes, cracks, ripples, lumps, excessive dust, discoloration, foreign matter, or any other defect which may cause undesirable effects during use.

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4.2.2 Inspection of preparation for delivery. An inspection shall be made to determine that the packaging, packing, and marking comply with the requirements of section 5. Defects shall be scored in accordance with table II. For examination of interior packaging the sample unit shall be one shipping container fully prepared for delivery, selected at random just prior to the closing operations. Sampling shall be in accordance with MIL-STD-105. Defects of closure listed shall be examined on shipping containers fully prepared for delivery. The lot size shall be the number of shipping containers in the end item inspection lot. The inspection level shall be S-2 with an AQL of 4.0 defects per hundred units.

TABLE II. Inspection of preparation for delivery defects.

Examine	Defects
Markings (exterior and interior)	Omitted; incorrect; illegible; improper size, location, sequence, or method of application.
Material	Any component missing or damaged.
Workmanship	Inadequate application of components such as incomplete closure of container flaps, loose strapping, or distortion of container.

4.2.3 Tests. Sample pieces of chalk shall be subjected to the following tests. Unless otherwise specified, tests shall be conducted in an atmosphere of 73.5 ± 2°F. with a relative humidity of 50 ± 4 percent. Sampling shall be in accordance with MIL-STD-105, inspection level S-2, with an AQL of 6.5 percent defective. The lot size shall be all packages containing one type and color of chalk offered for acceptance at one time, and the sample unit shall be one package. Failure of one or more pieces of chalk in a package to pass any test shall render the package defective.

4.2.3.1 Composition. To determine the amount of calcium sulfate in the chalk, use the applicable test procedure in ASTM Designation C471-61. Check for compliance with 3.2.1 or 3.2.2, as applicable.

4.2.3.2 Breaking strength, type I. Drop a sample piece of carpenters' chalk on a concrete floor from a height of 3 feet. Check for compliance with 3.6.

4.2.3.3 Breaking strength, type II. Select a sample piece of railroad chalk and place the piece perpendicularly across two horizontal parallel rods 1/4 inch in diameter and 2-1/2 inches apart. Position the chalk so that it is supported equally at both ends. Gradually apply an increasing load by means of a 1/4 inch diameter rod positioned midway between the ends of the chalk, perpendicular to the chalk piece and parallel to the supporting rods. Increase load up to a minimum of 3 pounds and check for compliance with 3.6.

5. PREPARATION FOR DELIVERY

5.1 Packaging. Packaging shall be level A, B or C as specified (see 6.2).

5.1.1 Level A.

5.1.1.1 Type I. Seventy-two (72) pieces of type I chalk of like color shall be packaged together in a close-fitting box conforming to PPP-B-636, W6s or W6c; PPP-B-676, type IV, grade B; PPP-B-665, style C or D; or a lock-corner wood box conforming to style 6 of PPP-B-621, except that one-piece sides, tops and bottoms not less than 3/16-inch thickness shall be used. If wood box is used, the cover shall be a slide type with a thumb notch for ease of opening. Suitable cushioning shall be provided by means of sawdust, shredded paper, or fiberboard or corrugated separators.

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5.1.1.2 Type II. One-hundred and forty-four (144) type II chalk of like color shall be packaged together in a close-fitting box conforming to PPP-B-636, W5s or W5c; PPP-B-676, type IV, grade B; PPP-B-665, style C or D; or a lock-corner wood box conforming to style 6 of PPP-B-621, except that one-piece sides, tops and bottoms not less than 3/16-inch thickness shall be used. If wood box is used, the cover shall be a slide type with a thumb notch for ease of opening. Suitable cushioning shall be provided by means of sawdust, shredded paper, or fiberboard or corrugated separators.

5.1.2 Level B.

5.1.2.1 Type I. Type I chalk of like color shall packaged as specified in 5.1.1.1 except the PPP-B-636 box shall be class-domestic, style optional.

5.1.2.2 Type II. Type II chalk of like color shall be packaged as specified in 5.1.1.2 except the PPP-B-636 box shall be class-domestic, style optional.

5.1.3 Level C. The chalk shall be packaged in accordance with the supplier's commercial practice.

5.2 Packing. Packing shall be level A, B, or C as specified (see 6.2).

5.2.1 Level A.

5.2.1.1 Type I. Ten type I unit packages of chalk of like description shall be packed in a close-fitting box conforming to PPP-B-636, class-weather resistant. The box shall be closed, waterproof sealed, and strapped in accordance with the appendix to the box specification.

5.2.1.2 Type II. Five type II unit packages of chalk of like color shall be packed in a close-fitting box conforming to PPP-B-636, class-weather resistant. The box shall be closed, waterproof sealed, and strapped in accordance with the appendix to the box specification.

5.2.1.3 Additional packing. When specified (see 6.2) unit packages of chalk of like description packed in the PPP-B-636 box shall be overpacked in close-fitting boxes conforming to PPP-B-585, class 3, PPP-B-591, class II, PPP-B-601, type III, PPP-B-621, class 2 or PPP-B-640, class 2, grade A. The boxes shall be closed and strapped in accordance with the applicable specification or appendix thereto. The gross weight shall not exceed 200 pounds. Strapping of the PPP-B-636 box will not be required when they are overpacked.

5.2.2 Level B.

5.2.2.1 Type I. Ten unit packages of type I chalk of like color shall be packed in a close-fitting box conforming to PPP-B-636, type CF, class-domestic. The box shall be closed in accordance with the appendix to the box specification.

5.2.2.2 Type II. Five unit packages of type II chalk of like color shall be packed in a close-fitting box conforming to PPP-B-636, type CF, class-domestic. The box shall be closed in accordance with the appendix to the box specification.

5.2.3 Level C. The chalk shall be packed to insure carrier acceptance and safe delivery to destination in containers complying with the rules and regulations applicable to the mode of transportation.

5.3 Standard Pack (Civil Agency Procurements). The Standard Pack for civil agency procurements shall be as specified in paragraphs 5.1.2 and 5.2.2 (see 6.3).

5.4 Marking.

5.4.1 Civil agencies. In addition to markings required by the contract or order, the packages and shipping containers shall be marked in accordance with Fed. Std. No. 123.

5.4.2 Military activities. In addition to markings required by the contract or order, the packages and shipping containers shall be marked in accordance with MIL-STD-129.

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6. NOTES

6.1 Intended use.

6.1.1 Type I. The carpenters' chalk covered by this specification is intended primarily for use in construction work for the purpose of transferring a mark from a length of string to the work.

6.1.2 Type II. The railroad chalk covered by this specification is intended primarily for the purpose of making indentification marks on wood, metal, stone, and concrete.

6.2 Ordering data. Purchasers should select the preferred options permitted herein and include the following information in procurement documents:

(a) Title, number, and date of this specification.

(b) Type required (see 1.2.1).

(c) Color required (see 1.2.2).

(d) If coating for type II is required (see 3.7).

(e) Selection of applicable level of packaging and packing required (see 5.1 and 5.2).

6.3 Standard Pack for civil agencies. The standard pack requirements in 5.3 are intended for use in procurements of Stores Stock replenishments. Procuring officers should use the standard pack requirements when it is known that the material will be shipped from a supplier to a domestic warehouse, supply depot, or intermediate storage point for temporary storage, subsequent issue, or shipment to eventual user.

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